Joint vertex covers and the envelope of Gowers's inverse induction

Formalization notes, Combinatorics/Ramsey (2026-10-11)

Abstract

In Gowers's proof of Szemerédi's theorem, the structural step of Section 16 (the single-piece lift) must produce one multilinear map that agrees with the *remainder* φ'' on a dense piece of a box. The remainder is a signed sum over the $2^k - 1$ non-top vertices of the cube $\{0,1\}^k$, and each vertex contributes a function that lives on a lower-dimensional face. If the vertices are handled one after another, each at the density left by the previous one, the degrees of the density losses multiply, and the resulting envelope grows like 3^k in the doubly-logarithmic scale. That outgrows the printed budgets 2^{d+10} (Theorem 18.1) and 2^{L+9} (Theorem 18.2) at degree 10 and length 14. We show that the vertex *covers* can be composed at a precision fixed in advance, with a single pigeonhole at the end. The losses then add, the envelope grows like 2^k, and Theorems 18.1, 18.2 and Corollary 18.7 hold in every degree and length, at the printed constants and with some room to spare. Every statement below is formalized in Lean (module names are given in §9).

1 Covers

Throughout, N is prime, a *box* is a product of arithmetic progressions in $\mathbb{Z}_N$ with a common difference, and $w(P)$ is the least side length of a box P. A map is *multilinear* in the sense of the paper.

Definition 1 (Cover)**.** Let $\Gamma \subseteq \mathbb{Z}_N^d \times \mathbb{Z}_N$ be a relation and $Q, E\colon (0,1] \to \mathbb{R}$. We say Γ is *covered with controls* (Q, E) if for every $\theta \in (0,1]$ and every proper box P there are

- a partition of P into proper boxes $Q_1, \dots, Q_M$ with $w(Q_j) \geq w(P)^{E(\theta)}$,
- a set $H \subseteq P$ with $|H| \geq (1-\theta)|P|$,
- for each j at most $Q(\theta)$ multilinear maps $\mu_{j,1}, \dots, \mu_{j,q}$,

such that every $(x, y) \in \Gamma$ with $x \in Q_j \cap H$ satisfies $y = \mu_{j,i}(x)$ for some i.

The precision θ is chosen by the user of the cover *before* it is applied, and the loss $\theta|P|$ is additive. A *provider*, by contrast, takes a set of density t and returns a piece of density $c(t)$; composing providers composes the functions c.

2 Joint covers

Lemma 2 (Joint cover)**.** *Let E be a finite index set, and for $e \in E$ let f_e be a function on a domain $D_e \subseteq \mathbb{Z}_N^d$ whose graph is covered with controls (Q_e, E_e), with $Q_e, E_e \geq 0$. Fix precisions $\theta_e \in (0,1]$. Then every proper box P has a partition into proper boxes $R_1, \dots, R_M$, a set $G \subseteq P$ and, for each j, at most $q \leq \prod_e Q_e(\theta_e)$ tuples $(\nu_{j,i,e})_{e \in E}$ of multilinear maps, such that*

$$|G| \geq \Big(1 - \sum_e \theta_e\Big)|P|, \qquad w(R_j) \geq w(P)^{\prod_e E_e(\theta_e)},$$

and every $x \in R_j \cap G$ lying in all domains D_e has an index i with $f_e(x) = \nu_{j,i,e}(x)$ for all e.

Proof. Induction on E. For $E = \emptyset$ take the trivial partition, $G = P$ and one empty tuple. For $E \cup \{a\}$, apply the induction hypothesis to P, obtaining cells Q_j, a good set G and tuples $\nu_{j,i}$. Apply the cover of f_a at precision θ_a to every cell Q_j; this gives sub-cells $R_{j,l}$, good sets whose union G' satisfies $|G'| \geq (1-\theta_a)|P|$ (the cells partition P), and maps $\mu_{j,l,i'}$. Take the sub-cells $R_{j,l}$ as the new partition, the good set $G \cap G'$, of size at least $|G| + |G'| - |P| \geq (1 - \sum_E \theta_e - \theta_a)|P|$, and the product tuples $(\nu_{j,i}, \mu_{j,l,i'})$. The width bound follows from $w(R_{j,l}) \geq w(Q_j)^{E_a} \geq w(P)^{E_a \prod_E E_e}$. □

Corollary 3 (One piece)**.** *In the setting of Lemma 2, put $s = \sum_e \theta_e$ and $Q = \prod_e Q_e(\theta_e)$. If $H \subseteq P$ lies in every domain D_e and $|H| \geq \theta|P|$ with $\theta > s$, then some cell R and some maps ν_e satisfy*

$$\big|\{x \in H \cap R : f_e(x) = \nu_e(x) \text{ for all } e\}\big| \geq \frac{\theta - s}{Q}\,|R|.$$

Proof. $|H \cap G| \geq (\theta - s)|P|$, so some cell R has $|H \cap G \cap R| \geq (\theta - s)|R|$. Each of its points is covered by one of at most Q tuples; the most popular tuple covers a $1/Q$ fraction. □

The only density loss is the single factor $1/Q$, and Q is a product of counts evaluated at *fixed* precisions. If $Q_e(\theta) \leq (C/\theta)^{D_e}$, the density has degree $1 + \sum_e D_e$ in $1/\theta$: the degrees add.

3 The remainder of the single-piece step

On the good domain of the step, the remainder is $\varphi''(x) = -\sum_{e \neq \mathbf{1}} (-1)^{k+|e|} \varphi_e(x)$, where φ_e is the translated cube vertex e, a pullback of φ to the face spanned by the directions S_e, of dimension $|e| + 1$. By induction on the dimension, the face carries a cover of degree $D_{|S_e|}$. It lifts to the ambient box without any loss in the precision: the count becomes $\max(Q, 3^{k+1})$ and the exponent $E/16^{k+1-|S_e|}$. Applying Corollary 3 with $\theta_e = \theta_2/2^{k+1}$ to the dense good set (density θ_2) gives one map $M'' = -\sum_e (-1)^{k+|e|} \nu_e$ on a piece of density

$$\rho_1 = \frac{\theta_2/2}{\prod_{e \neq \mathbf{1}} \max\big(Q_{|S_e|}(\theta_2/2^{k+1}), 3^{k+1}\big)}$$

and width at least $N^{\prod_e E_e/16^{k+1-|S_e|}}$.

Remark 4 (Nesting versus joint covers). With the class degrees $D_l = 2^{68 \cdot 2^l}$, nesting one provider per vertex gives $\log_2 \deg \rho_1^{-1} \approx \sum_e 68 \cdot 2^{|e|+1} = 136 \cdot 3^k$. The joint cover gives $\log_2\big(1 + \sum_e D_{|S_e|}\big) \leq 68 \cdot 2^k + k$.

4 The envelope

Write $X = 2/\alpha$. The inverse induction carries bounds $\beta^{-1}, \sigma^{-1} \leq X^{2^\mu}$ from degree $d-1$ to degree d. With the joint-cover remainder, one step in dimension k costs

$$\mu_{\text{new}} = \mu + \underbrace{(68 \cdot 2^k + \mu_0 + k + 3)}_{\rho_1} + 2 \cdot 68 \cdot 2^k + O(k^2),$$

where μ_0 bounds the base budget of the cover controls. With the standard $\mu_0 \approx 64 \cdot 2^k$ one step costs $268 \cdot 2^k + O(k^2)$. Hence

Theorem 5. *For every $k \geq 1$ the Theorem 18.1 envelope satisfies $\mu_k \leq 1000 \cdot 2^k$, and the Theorem 18.2 envelope (which also carries thresholds) satisfies $\mu_k \leq 1000 \cdot 2^k$ as well. Both are well inside the printed budgets $2^{d+10} = 4096 \cdot 2^{d-2}$ and $2^{L+9} = 8192 \cdot 2^{L-4}$. Consequently Theorems 18.1 and 18.2 and Corollary 18.7 hold in every degree and length.*

5 Sharper constants

The same proofs give smaller tower constants. Moreover, the class base $V = 1 + \sum_{j<k}(K_j + p_j)$ satisfies $V \leq 2^{2^{2^{k+4}+1}}$, so the base budget $\mu_0 = 2^{k+5} + 9$ suffices; one step then costs $236 \cdot 2^k + O(k^2)$ and the envelopes grow like $472 \cdot 2^k < 512 \cdot 2^k$.

	printed	standard base	tight base	Fejér θ_1 (§6)
Theorem 18.1, exponent $\alpha^{2^{2^{d+c}}}$	$c = 10$	$c = 8$ ($d \geq 4$)	$c = 7$	$c = 6$ (every d)
Theorem 18.2, threshold with $L + c$	$c = 9$	$c = 6$ (every L)	$c = 5$	$c = 4$ (every L)
Corollary 18.7, threshold with $L + c$	$c = 9$	$c = 6$ (every L)	$c = 5$	$c = 4$ (every L)
$W(k, r) \leq 2^{2^{r^{2^{2^{k+c}}}}}$, $r \geq 2$	$c = 9$	—	$c = 5$	$c = 4$
Theorem 1.3, density $(\log\log N)^{-2^{-2^{k+c}}}$	$c = 9$	—	$c = 5$	$c = 4$

The $W(k, r)$ row is Theorem 18.2 at density $\delta = 1/r$: in an r-colouring of $[1, N]$ one colour class has at least N/r elements. This gives Gowers's van der Waerden bound, sharpened from $k + 9$ to $k + 5$. The headline Theorem 1.3 follows from Theorem 18.2 by a calculation in which the constant enters only through 2^{k+9}; repeated with $k + 5$, it gives the density exponent $2^{-2^{k+5}}$.

With the printed θ_1 the method stops at the tight-base constants. The envelope recursion grows like $472 \cdot 2^k$, and $c = 6$ for Theorem 18.1 (or $c = 4$ for Theorem 18.2) needs a step cost below $128 \cdot 2^k$, while the vertex covers and the two slice controls alone cost $204 \cdot 2^k$. The last column of the table comes from changing θ_1 itself (§6).

For Theorem 18.2 at $c = 5$ one more point matters. The modulus threshold of the inverse step contains two pure constants, the Section 5 refinement constant $\leq 2^{2^{40d^3+2}}$ and a boundary constant $\leq 2^{2^{1083}}$. A cruder bound charges them at the level of powers of X, which puts a cubic term $40(k+2)^3$ into the envelope and blocks $L + 5$ for $L \leq 10$. The threshold is only needed below $\exp(\exp(X^{2^\mu}))$, and $2^{2^m} \leq \exp(X^{2^j})$ as soon as $m \leq 2^j$. At that level the constants cost only $k + 16$, and $L + 5$ holds in every length.

6 A Fejér selection and the constants $d + 6$, $L + 4$

The class degree $68 \cdot 2^l$ and the base budget both come from the Section 16 density $\theta_1 = (\theta\gamma/4)^{2^{2^{k+5}}}$. Its double exponential is created in Lemma 15.5. A point is kept with probability $(1+\cos)/2$, which gives a respected arrangement of $M = 2^{k+4}$ vertices an advantage of only $1 + 2^{1-M}$ per stage, so 2^{M-1} stages are needed.

One Fejér stage. Keep a point with the Fejér weight $F_m(u) = |\sum_{j \leq m} e(ju/N)|^2/(m+1)^2$ of order $m + 1 = M\lceil 6/(\alpha\eta)\rceil$. Averaged over the coefficients, the survival probability of an arrangement is a sum over the integer vectors $\eta \in [-m, m]^M$ annihilating all its monomials. For arrangements that are not m-degenerate, these are the multiples $j\varepsilon$ of the parity vector. They contribute $\sum_{|j| \leq m}(1 - |j|/(m+1))^M \geq T$ for respected arrangements and only $j = 0$ otherwise. Lemma 15.4 extends to coefficients in $[-m, m]$, since Corollary 15.2 holds for every coefficient

vector, and it bounds the m-degenerate arrangements. The resulting Lemma 15.5 has exponent $2^{k+4}(k+6)$, and Lemma 15.6 has

$$\theta_1' = (\theta\gamma/4)^F, \qquad F \le 2^{5k+14},$$

in place of $2^{2^{k+5}}$.

The tight radius. The Bohr radius $\zeta = 2^{-s}$ only has to satisfy Lemma 16.5's condition, which needs $s \ge 2^{83}/(\theta_1/4)^{11}$. With θ_1' this is $s' \le (2/\theta\gamma)^{22F+105}$, instead of the printed $(2/\theta\gamma)^{2^{2^{k+6}}}$.

Propagation. With these inputs the dimension-two class has degree 2^{46} (printed 2^{132}). The split-class recursion keeps $\log_2 D_l \le 22 \cdot 2^l$, so the class covers have degree $2^{23\cdot 2^l}$ in place of $2^{68\cdot 2^l}$. Since $\theta_1' \ge \theta_1$, the budget, θ_2, δ and the radius of Section 18 are at least the printed ones, and every printed fixed-parameter bound transfers unchanged. One step of the inverse induction then costs

$$\mu_{\text{new}} = \mu + \underbrace{23\cdot 2^k}_{\text{vertices}} + \underbrace{32\cdot 2^k}_{\mu_0} + \underbrace{2\cdot 23\cdot 2^k}_{\text{slices}} + O(k^2) = \mu + 101\cdot 2^k + O(k^2),$$

and both envelopes are at most $230\cdot 2^k$ for $k \ge 4$, below the $256\cdot 2^k$ of $c = 6$ and $c = 4$.

Theorem 6. *Theorem 18.1 holds in every degree d with exponent $\alpha^{2^{2^{d+6}}}$, and Theorem 18.2 and Corollary 18.7 hold in every length L with $L+4$ in the threshold. Hence $W(k,r) \le 2^{2^{r^{2^{2^{k+4}}}}}$, and Theorem 1.3 holds with density exponent $2^{-2^{k+4}}$.*

Large degrees. The proved class bound $23\cdot 2^l$ is loose. The class recursion only needs slack that grows additively, and indeed

$$2^{l+6} a_l \le 2^{14\cdot 2^l}, \qquad D_l \le 2^{14\cdot 2^l + 3(l+4)^2}$$

in every dimension $l \ge 2$, since $3(l+5)^2 - 3(l+4)^2 = 6l + 27$ covers the $5l + 22$ that one step adds. Moreover the Fejér θ_1 no longer forces the base budget $32\cdot 2^k$: the class-base term $16\cdot 2^k$ suffices. One step then costs $58\cdot 2^k + O(k^2)$, and the envelopes stay below $128\cdot 2^k$ from $k = 10$ on.

Theorem 7. *Theorem 18.1 holds with exponent $\alpha^{2^{2^{d+5}}}$ at every degree $d \ge 12$, and Theorem 18.2 and Corollary 18.7 hold with $L+3$ at every length $L \ge 14$; in particular $W(k,r) \le 2^{2^{r^{2^{2^{k+3}}}}}$ for $k \ge 14$, and Theorem 1.3 holds with density exponent $2^{-2^{k+3}}$ for $k \ge 14$.*

The small cases stay at $d + 6$ and $L + 4$ because of the class recursion's transient near $l = 4$, where the family exponent squares, $a_{l+1} \approx 2^{l+5} a_l^2$.

The dimension-two family exponent. Since the family exponent squares, $\log_2 a_l \approx (\log_2 a_2/4)\, 2^l$, so the constant 14 above is set by $a_2 = 2^{46}$. That value only reflects the passage from a power class, which sets the family exponent equal to the parameter degree. The dimension-two controls are polynomials of fixed degree in the number $M = n\,m$ of pieces: the base controls are linear in M. In the two bases $\nu = n + 1$ and $W = 2V/(\gamma\theta\rho)$:

- the count is at most $\nu^{11} W^{2^{41}}$;
- the inverse width exponent is at most $\nu^{29} W^{2^{46}}$.

So dimension two is a split class with $a_2 = 32$ and $D_2 = 2^{48}$, and the invariants become

$$2^{l+6} a_l \le 2^{4\cdot 2^l}, \qquad D_l \le 2^{4\cdot 2^l + 3(l+4)^2 + 30}.$$

The class degree is then $4 \cdot 2^l + 3(l+4)^2 + 32$.

At this class degree the density degree of the step is about $28 \cdot 2^k$. That is below the $32 \cdot 2^k$ which the η step of the printed envelope asks of it, but the η step only absorbs the constant $2^{2(k+2)^3}\, 2^{k+1}(2/\alpha)^2$, and $3(k+4)^2$ suffices for that. One step then costs $28 \cdot 2^k + O(k^2)$, so the Theorem 18.1 envelope tends to $56 \cdot 2^k < 64 \cdot 2^k$.

Theorem 8. *Theorem 18.1 holds with exponent $\alpha^{2^{2^{d+4}}}$ at every degree $d \geq 13$, and with $\alpha^{2^{2^{d+5}}}$ at every degree $d \geq 9$. Theorem 18.2 and Corollary 18.7 hold with $L+3$ at every length $L \geq 11$; in particular $W(k,r) \leq 2^{2^{r^{2^{2^{k+3}}}}}$ and Theorem 1.3 holds with density exponent $2^{-2^{k+3}}$ for $k \geq 11$.*

The Theorem 18.2 envelope cannot reach $L+2$ in this way: its threshold branch alone is $64 \cdot 2^k + 8 > 64 \cdot 2^k$. Nor can $d+3$ be reached, since the base budget contributes $16 \cdot 2^k$ to every step.

7 Theorem 16.2 and Corollary 16.11 with polynomial controls

The printed Theorem 16.2 restricts a relation with the product property to a set of density $1-\theta$ on which it is multiply linear, with graph count $q(r^{-1}\rho, \gamma, k)^r$ at precision ρ, where $r = \gamma^{-2} s(\theta, \gamma, k)$ and $s = (2/\theta\gamma)^{2^{2^{k+6}}}$. This count is exponential in a polynomial of $1/\theta$. The covers built for the dimension induction give a count that is *polynomial* in $1/(\gamma\theta\rho)$:

$$Q(\gamma, \theta, \rho) = \max\Big(1, \Big(\frac{4V}{\gamma\theta\rho}\Big)^{2^{68 \cdot 2^k}}\Big), \qquad V = 2^{2^{2^{k+4}+1}},$$

with width exponent Q^{-1}, in every dimension $k \geq 1$ (`theorem_16_2_polynomial`). This form implies the printed one (`theorem_16_2_polynomial_implies_printed`).

The same holds for Corollary 16.11. For a function that is not uniform of degree $k+2$, the single-piece step with the joint-cover remainder gives a box of dimension $k+1$, of width N^e, on which a multilinear frequency is large on an e fraction, with

$$e^{-1} \leq (2/\alpha)^{2^{236 \cdot 2^k + k + 24}}$$

for every $k \geq 2$ (`corollary_16_11_polynomial`). The printed exponent is $(\alpha/8)\, c(r^{-1}\alpha/8, \alpha/2)^r$ with r polynomial in $1/\alpha$, so its reciprocal is exponential in a polynomial of $1/\alpha$.

With the Fejér classes of §6 both exponents shrink:

- the graph count becomes $\max(1, (4V/\gamma\theta\rho)^{2^{23 \cdot 2^k}})$ (`theorem_16_2_polynomial_fejer`);
- the corollary has $e^{-1} \leq (2/\alpha)^{2^{101 \cdot 2^k + k + 24}}$ (`corollary_16_11_polynomial_fejer`).

With the dimension-two family exponent 32 (Theorem 8) they shrink further:

- the graph count becomes $\max(1, (4V/\gamma\theta\rho)^{2^{4 \cdot 2^k + 3(k+4)^2 + 32}})$ (`theorem_16_2_polynomial_fejerS`);
- the corollary has $e^{-1} \leq (2/\alpha)^{2^{32 \cdot 2^k + 9(k+4)^2 + k + 120}}$ (`corollary_16_11_polynomial_fejerS`).

8 Further questions

1. *Joint slices.* The single-piece step applies its slice control twice in succession, $\theta\, c(c(\theta))$, and so pays $2 \cdot 68 \cdot 2^k$ in the step cost. Composing the two slice covers jointly, as in Lemma 2, should cost about $68 \cdot 2^k$. The envelope would drop from about $472 \cdot 2^k$ to about $336 \cdot 2^k$. This changes none of the constants in the table.

2. *Small degrees.* With the dimension-two family exponent 32 (Theorem 8) the class constant is 4, and $c = 5$ holds from degree 9 on and $c = 4$ from degree 13 on. Below degree 9 the obstacle is the additive slack $3(l+4)^2 + 32$ of the class degree, which dominates $4 \cdot 2^l$ for small l. The Theorem 18.1 envelope peaks at about $212 \cdot 2^k$ at $k = 3$. Is there a class recursion whose slack is linear in l, or whose small dimensions are bounded directly? That would carry $c = 5$ to every degree.

3. *Fixed-precision slices with the Fejér classes.* Composing the two slice controls jointly would lower the step cost from $101 \cdot 2^k$ to about $78 \cdot 2^k$. That is still above $64 \cdot 2^k$, so it changes no constant in the table.

9 Formalization

All statements are in the directory `Combinatorics/Ramsey/Lean/GowersSzemeredi`.

Statement	Lean
Lemma 2, Corollary 3	`joint_cover_on`, `JointCoverOn.piece` (`Proofs16JointCovers`)
the remainder piece	`remainder_piece_of_vertex_covers`
the single-piece step	`single_piece_lift_global_cover` (`Proofs16SinglePieceCoverGlobal`)
ρ_1 bound	`rho1_inv_leC` (`Proofs18CoverDensity`)
envelopes	`polyGenMuC_le`, `genMuC_le`, `genMuC_le_sharp`
Theorems 18.1, 18.2, Cor. 18.7	`theorem_18_1_explicit`, `theorem_18_2_explicit`, `corollary_18_7_explicit` (`Proofs18CoverExplicit`)
sharper constants	`Proofs18CoverSharper`, `Proofs18CoverTightBase`, `Proofs18CoverColorings`, `Proofs18CoverSharperExplicit`, `Proofs18CoverSharperLowDegrees`, `Proofs18CoverSharperShortLengths`, `Proofs18CoverThresholdsSharp`, `Proofs18CoverVanDerWaerden`, `Proofs18CoverHeadline`, `Proofs18CoverSharperAllDegrees`
Fejér Lemmas 15.5, 15.6	`Proofs15FejerSelection`, `Proofs15FejerProduct`
tight radius, Fejér classes	`Proofs16TightRadius`, `Proofs16UniformCoverTwoFejer`, `Proofs16ClassStepDegreesFejer`, `Proofs16SplitStepFejer`
Fejér envelope	`Proofs18TightClassesFejer`, `Proofs18FejerEnvelopeBase`, `Proofs18CoverFejerHeadline`
Theorem 6	`theorem_18_1_d6`, `theorem_18_2_L4`, `corollary_18_7_L4`, `van_der_waerden_L4`, `theorem_1_3_L4` (`Proofs18CoverFejerAllDegrees`)
Theorem 7	`theorem_18_1_d5_twelve_le`, `theorem_18_2_L3_fourteen_le`, `Proofs18TightClassesFejerQ`, `Proofs18CoverFejerHeadlineQ`
Theorem 8	`theorem_18_1_d4_thirteen_le`, `theorem_18_1_d5_nine_le`, `theorem_18_2_L3_eleven_le`, `theorem_1_3_L3_eleven_le`, `Proofs16UniformCoverTwoSplit`, `Proofs18TightClassesFejerS`, `Proofs18EtaStepSmall`, `Proofs18CoverFejerHeadlineS`
Fejér polynomial 16.2, 16.11	`Proofs16PolynomialFejer`, `theorem_16_2_polynomial_fejer`, `corollary_16_11_polynomial_fejer`
polynomial Theorem 16.2	`Proofs16PolynomialStructure`, `theorem_16_2_polynomial`
polynomial Cor. 16.11	`Proofs16PolynomialCorollary`, `corollary_16_11_polynomial`

Every result above is kernel-checked. The production audit of 2026-10-11,

`check_gowers.py GowersSzemeredi.CompletionAudit`,

built 1,440 modules from source with Lean 4.32.0, checked the exact types of all 120 numbered catalogue statements and of the improved-bound theorems, and found only the axioms `propext`, `Classical.choice` and `Quot.sound`. The extensions to every degree ($c = 7$) and to every length ($c = 6$ and $c = 5$), the van der Waerden and Theorem 1.3 rows, and the polynomial Theorem 16.2 and Corollary 16.11 postdate that audit. They are so far checked against faithful stubs, and `CompletionAudit.lean` now includes them. The audit's record is

`Research/GowersSzemeredi/verification/2026-10-11-numbered-and-bounds.json`.

The Fejér column (§6, 2026-10-12) is newer still. Its light modules are kernel-checked locally with the standard axioms only. Its bridge `Proofs18CoverFejerAllDegrees` is checked against a faithful stub whose axioms are exactly those of the $c = 7/c = 5$ bridge, namely the corpus

theorems that the audit verified. It is not yet in `CompletionAudit.lean`. The research notes (`SECTION16_RESEARCH_NOTES.md`, §§L.11–L.16e) record the development.